DATA SCIENCE LABORATORY MANUAL

Module I ? Data Science Fundamentals

Topic: Data Collection

OVERVIEW & LEARNING OBJECTIVES

Data ingestion through web scraping, REST APIs, SQL databases, sensor streams, and public datasets.

KEY TOPIC CONCEPTS & METHODOLOGY

- Understanding Data Collection in modern Data Science workflows.
- Detailed breakdown: Data ingestion through web scraping, REST APIs, SQL databases, sensor streams, and public datasets
- Mathematical formulas, code examples, and laboratory execution guidelines.
- Common pitfalls, edge cases, and best practices in practical applications.

LABORATORY PRACTICE / IMPLEMENTATION NOTES

1. Follow standard Python and Data Science coding practices.
2. Document all exploratory findings and statistical hypotheses.
3. Keep datasets and outputs organized in your repository workspace.